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PALEOMAGNETISM OF BASALTS FROM ALBORZ: IRAN PART OF ASIA IN THE CRETACEOUS

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ABSTRACT

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Paleomagnetic results are reported from 20 sites within three units of volcanic rocks of Cretaceous age from the Central Alborz Mountains, Iran. After application of progressive demagnetization either with alternating magnetic fields or with heating, the mean characteristic remanence direction is found to be $D = 33.3^\circ$, $I = 47.5^\circ$ ($\alpha_{95} = 7.2^\circ$). Both the paleomagnetic analyses and the inspection of polished thin sections of the lavas point to magnetite as the main carrier of remanence. The paleomagnetic pole is located at 61°N, 147.5°E. This is quite near to the positions of the poles of Cretaceous age on the Eurasian apparent polar wander path. It is suggested that in Cretaceous times northern Iran was at its present position with respect to Eurasia.

INTRODUCTION

The central part of the Alborz Mountains is usually subdivided into seven structural units, viz., from north to south: (1) Caspian Plain; (2) Northern Mesozoic Border Zone; (3) Paleozoic Central Ranges; (4) Central Tertiary Zone; (5, 6) Southern Paleozoic—Mesozoic Zone; (7) Southern Frontal Depression (Gansser and Huber, 1962; Stöcklin, 1968). In this orogen the main tectonic phenomena are blocks bounded by faults; the sediments within these blocks are often strongly folded, but no large thrusts are encountered.

For our paleomagnetic research we have collected oriented samples from the volcanics of Cretaceous age (Fig. 1) both in the Northern Mesozoic Border Zone (unit 2) and in the Southern Paleozoic—Mesozoic Zone (unit 5). In unit 2, samples were collected from the Chalus Formation in the valley of the river Chalus (Fig. 2); in unit 5, volcanic rocks were sampled from the Gypsum — Melaphyr Formation in the valley of the river Haraz, east of Mount Damavand.

The aim of this paleomagnetic study is to determine the characteristic directions of remanent magnetization of these rocks, and the significance for the paleoposition of northern Iran.

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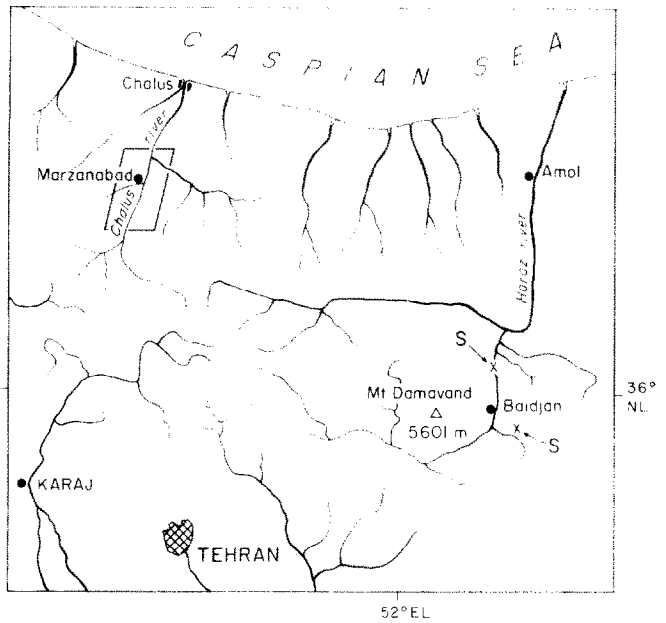


Fig. 1. Map of the central part of the Alborz Mountains, Iran, with the sampling localities of the volcanics of the Gypsum–Melaphyr Formation, and the index map of the Chalus area of Fig. 2.

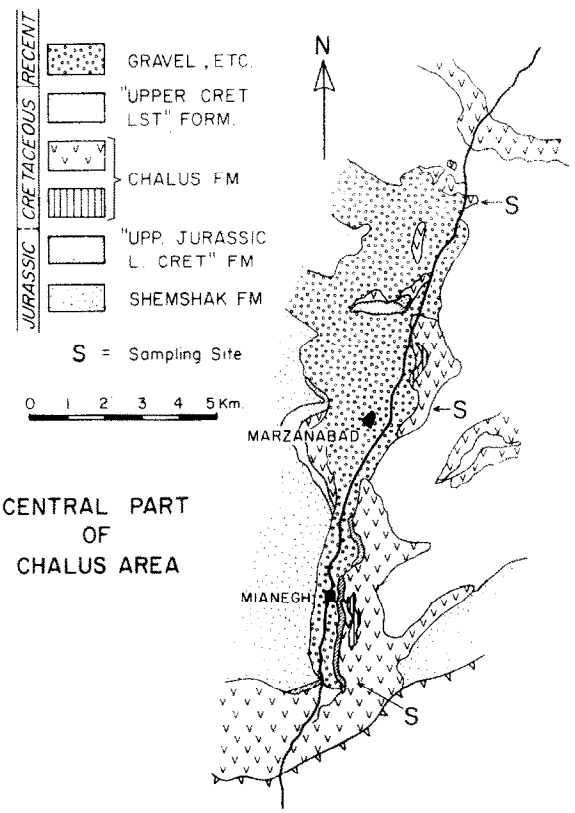


Fig. 2. Geological map of the central part of the Chalus area with the sampling localities of the volcanics of the Chalus Formation. (After Cartier, 1971.)

GEOLOGICAL PART

In the area of the present Alborz Mountains strong epeirogenetic movements occurred following the deposition of various kinds of sediments of Triassic age, which resulted in the emersion of the region. The subsequent erosion caused the deposition of detrital rocks of the Shemshak Formation. These rocks consist of grey sandstones, conglomerates and shales occasionally with intercalations of coal. The Shemshak Formation, which has an Early to Middle Jurassic age, covers an extensive area, and in the Alborz has a thickness ranging from about 1000 m in the south to about 2500 m in the north.

In the Central Alborz, the overlying sediments of Late Jurassic and Cretaceous age show considerable differences in facies development (Fig. 3).

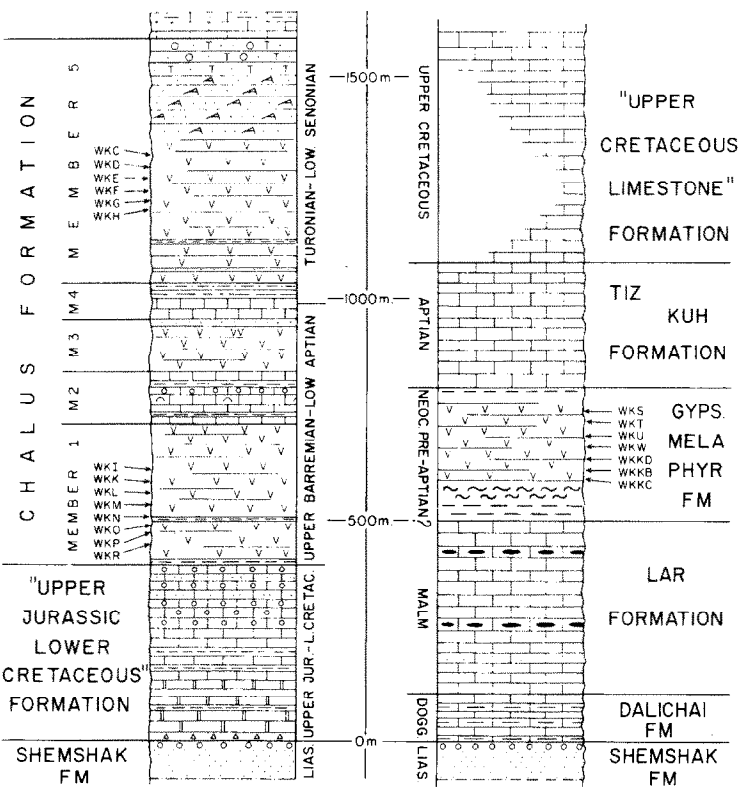


Fig. 3. Stratigraphic columns of the formations in the central part of the Alborz Mountains from Early Jurassic through late Cretaceous age in the Chalus area (left) and in the area east of Mount Damavand (right), with the positions of the sites. The positions of the sites WKA and WKB from member 1 of the Chalus Formation are not known exactly, they are not indicated in the column. (After Allenbach, 1966; Cartier, 1971; Steiger, 1966.)

In the area east of Mount Damavand marly fossiliferous limestones are found on top of the Shemshak Formation. The limestones are about 100 m thick and have a Middle Jurassic age. The succeeding Lar Formation consists of 400 m of pure, fossiliferous limestones with ammonites, pointing to an Late Jurassic age. In the Chalus valley, about 80 km northwest of Mount Damavand, there are no sediments of Middle Jurassic and early Late Jurassic age. In the northern part of the Chalus area, sedimentation started with transgression conglomerates and breccias overlain by dolomites with intercalations of red marl, red silts, and reddish sandstones; hereupon follow limestones with biosparites and oösparites with intercalations of reddish sandy limestones. In the southern part of the Chalus valley, sedimentation begins later, and here the sequence consists of limestones with intercalations of reddish sandy beds. The thickness of the formation decreases from 400 m in the north to 100 m in the south. The microfauna points to an age ranging from Late Jurassic to Middle Barremian age (Cartier, 1971).

During Early Cretaceous times there was a regression in the area northeast to southeast of Mount Damavand. After the deposition of the Late Jurassic limestones of the Lar Formation, layers of gypsum were laid down, followed by dark-coloured subaerial lava flows — mainly basalts — with a total thickness of up to 200 m. The top layer often consists of a paleosoil with crusts of iron oxyde (Steiger, 1966). The age of this formation, the Gypsum—Melaphyr Formation, is post-Malm and pre-Aptian. Then follow approximately 250 m of light-coloured Orbitolina limestones of the Tiz Kuh Formation, which indicates marine conditions during Aptian times. There are no sediments of Albian age. Marine conditions prevailed during most of the Late Cretaceous.

During Cretaceous times, in the Chalus area different conditions prevailed. In the period post-Malm to pre-Barremian there are neither deposits of gypsum or volcanics. We have seen that during Late Jurassic and Early Cretaceous times limestones with sandy intercalations were deposited. Hereupon follows the Chalus Formation with a very thick sequence of volcanics alternating with limestones. This formation, the type section of which is about 1200 m thick, is subdivided into five members. The successive members have markedly different thicknesses. Member 1 is 320 m thick in the type section, and is built up of many dark-coloured, basic lavas with thin intercalations of tuffite. Member 2 is about 100 m thick, and consists of variegated limestones with sandy and silty intercalations; the Orbitolines indicate a Late Barremian to Aptian age. The next member is composed of basic lava flows and is up to 100 m thick. Member 4 is only 40 m thick. Its lower part consists of limestones with Orbitolines, indicating a Late Barremian and Early Aptian age. After a break in the sedimentation, sandy limestones and shales follow; these littoral sediments with macrofossils have a Turonian to Early Senonian age. Member 5 is up to 550 m thick, and is built up of lavas and agglomerates. The Chalus Formation is overlain by thick-bedded limestones and marls with abundant *Globotruncana*, pointing to a Late Cretaceous age.

SAMPLING LOCALITIES

For our paleomagnetic research we collected oriented samples from three volcanic units (Table I). The collection consisted mainly of hand samples. In the mountainous area portable drilling equipment could be used sporadically. In the area near the river Haraz east of Mount Damavand, the volcanics of the Gypsum—Melaphyr Formation were sampled at two localities: one from four lava flows in the mountains 2 km southwest of Lut, and the other from three lava flows in the Haraz valley, 10 km north of Baidjan along the road from Tehran to Amol (Figs. 1 and 3). In the Chalus area we sampled both member 1 and member 5 from the Chalus Formation. Samples were collected from member 1 at two localities. The first locality included two sites 17 km south of Chalus, one being a lava flow and the other a tuffite (WKB). The second locality, which was further south, 3 km south of Mianegh, included seven lava flows and one tuffite (WKN). Member 5 was sampled east of Marzanabad at six sites representing six successive lava flows (Figs. 2 and 3). In all, 103 hand samples and 30 cores were collected from 21 lava flows and 2 tuffite layers.

PALEOMAGNETIC TREATMENT

In the laboratory, cores with a diameter of 25 mm were drilled from the hand samples. All cores were cut into segments 22 mm long. In Table I, we list for each site both the number of hand samples or cores and the number of specimens used in the analyses. For the paleomagnetic research use was made of the standard equipment available. The procedures were as follows. From each site one pilot specimen was subjected to a progressive demagnetization in alternating magnetic fields (af) in 12—15 successive steps up to 200 mT peak (2000 Oe) and occasionally even up to 300 mT peak value (3000 Oe). Eighteen specimens, selected from individual sites, were progressively heated up to 574° C in 9 successive steps in an argon-filled furnace.

After each step of progressive demagnetization the specimens were measured either on an astatic or a spinner magnetometer. The progressive partial demagnetization can be visualized by means of a graphical method in orthogonal projection. This method often works because normally the coercivity spectra of the carriers of the differently directed magnetic remanences exhibit significantly wide windows free of overlap (Zijderveld, 1975, Roy and Lapointe, 1978). The method shows the degree of isolation of the characteristic remanence direction; in a particular specimen this direction has been obtained if the vector of remanence is a straight line, which is directed towards the centre of the coordinate system and which decreases in length only without changing its direction during the subsequent steps of progressive demagnetization.

The main pilot specimens of those sites, the one specimen treated with af and the other with heating, were drilled from the same hand sample so that

TABLE I

Paleomagnetic data of rocks of Cretaceous age from the Alborz Mountains, Iran

Locality	Site	Attitude	Age	T	*S*	*E*	*N*	*D*	*I*	*k*	α_{95}
Chalus Fm — member 5											
Chalus	WKC	N43E 30	Tur.—Con.	af(6), th(1)	5	—	7	35	50.3	109	5.8
valley E of	WKD	N43E 30	Tur.—Con.	af(10), th(1)	8	—	11	38.7	56.8	119	4.2
Marzanabad	WKE *	N43E 30	Tur.—Con.	af(9), th(1)	7	—	5	65.7	21.6	172	5.8
						—	5	25.1	63.2	238	5.0
	WKF	N43E 30	Tur.—Con.	af(2), th(1)	6	3	—	—	—	—	—
	WKG	N33E 30	Tur.—Con.	af(9), th(1)	6	1	9	31.2	41.2	29	10.4
	WKH	N38E 27	Tur.—Con.	af(9), th(1)	7	—	9	36	46.3	81	5.7
Chalus Fm. — member 1											
Chalus	WKI	N18E 20	Barr.—Apt.	af(8), th(1)	5	—	9	19	48.1	103	5.1
Valley 3 km	WKK	N18E 20	Barr.—Apt.	af(7), th(1)	5	—	8	19.7	58	63	7.0
of Mianegh	WKL	N18E 20	Barr.—Apt.	af(4), th(1)	4	1	4	18.6	51.9	125	8.2
	WKM	N18E 20	Barr.—Apt.	af(8), th(1)	5	—	9	29.6	45.5	108	5.0
	WKN	N18E 20	Barr.—Apt.	af(5), th(1)	5	—	6	68.2	—0.5	38	11.1
	WKO *	N21E 18	Barr.—Apt.	af(7), th(1)	6	—	3	40.4	51.9	111	11.8
						—	5	177.5	—63.3	123	6.9
	WKP	N21E 18	Barr.—Apt.	af(5), th(1)	5	1	5	168.9	—43.4	52	10.7
	WKR	N21E 18	Barr.—Apt.	af(5), th(1)	4	—	6	198.1	—40.7	162	5.3
5 km N of	WKA	S29E 50	Barr.—Apt.	af(7), th(1)	(8)	1	7	30.8	31.4	65	7.6
Marzanabad	WKB	S29E 50	Barr.—Apt.	af(6), th(1)	5	—	7	218.2	—59.3	106	5.9
Gypsum-Melaphyr Fm.											
Haraz valley	WKS	N72W36	post-Malm	af(6)	5	—	6	49.6	29.2	41	10.6
SSW of Lut	WKT	N72W36	post-Malm	af(1), th(1)	5	2	—	—	—	—	—
	WKU	N72W36	post-Malm	af(6)	6	1	5	250.4	—41.1	87	8.2
	WKW	N72W36	—	af(6), th(1)	4	—	7	247.9	—55.4	198	4.3
10 km N of	WKKD	N89W96	pre-Aptian	af(7)	(7)	2	5	178.7	—54.4	20	17.5
Baidjan	WKKB	N85W88	pre-Aptian	af(7)	(8)	—	7	17.6	41.2	45	10
	WKKC	N80W85	pre-Aptian	af(8)	(7)	—	8	37.5	43.5	72	6.6

T is the demagnetization procedure applied: af and th are alternating magnetic fields and heating, respectively, the number of specimens treated is in brackets; *S* is the number of handsamples or (cores); *E* and *N* are the number of specimens excluded from and included in the ultimate analysis, respectively. *D* and *I* are the declination and inclination in degrees of the characteristic magnetization direction after correction of tilt; *k* is the precision parameter; α_{95} is the semi-angle of the cone of 95% confidence, in degrees.

* Site with separately clustered directions.

the results of both demagnetization procedures could be compared. Apart from these pilot specimens, usually a few more specimens from a particular site have been progressively demagnetized with af in about 10 successive steps up to at least 100 mT peak value (1000 Oe). Af demagnetization was applied to the remaining specimens in 4—5 successive steps.

GYPSUM—MELAPHYR FORMATION

Progressive af demagnetization of pilot specimens from all seven sites and progressive heating of two specimens show that the characteristic directions of magnetization are usually obtained after application of fields of 40 mT and heating at about 400°C, respectively (Fig. 4). The decay of the remanence intensity during progressive demagnetizations is illustrated in Fig. 5 (WKW 4A and WKW 4). The decay curves show that at first the remanence intensities increase slightly. From the demagnetization diagrams of Fig. 4 it can be inferred that the resultant vector of remanence increases in length during progressive demagnetization up to 40 mT and 237°C, respectively, i.e., during the decay of the secondary component of magnetization, the direction of which strongly deviates from that of the characteristic component.

The combined analysis shows that the Gypsum—Melaphyr Formation has both normally (N) and reversely (R) magnetized lavas (Table I). One lava flow — site WKT — has a characteristic magnetization direction that strongly deviates from the mean value for this formation: this datum is discarded in the ultimate analysis. The large dispersion in the magnetization directions may be the result of the application of inaccurate data for the tectonic correction to some lava flows near Lut — besides WKT, also sites WKU and

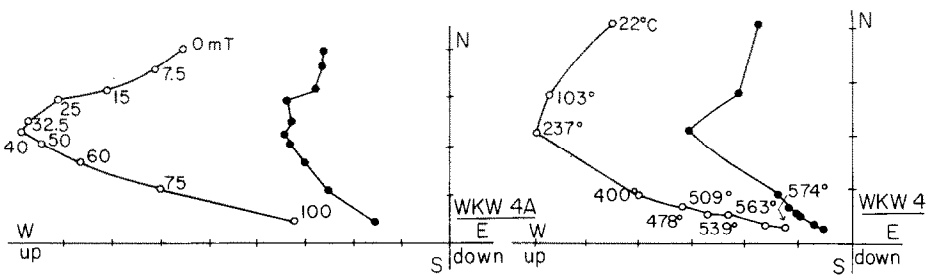


Fig. 4. Diagrams showing the progressive demagnetization of specimens from site WKW of the Gypsum—Melaphyr Formation with alternating magnetic fields (left) and with heating (right) before tectonic correction. The plotted points represent successive positions in an orthogonal projection of the end of the resultant NRM vector during progressive demagnetization. Solid and open circles denote the projections on a horizontal and on a north—south vertical plane, respectively. The numbers represent the peak strength in mT (1 mT = 10 Oe) of af (left) and the temperature applied in degrees centrigade (right). On both diagrams each unit on either axis represents 0.1 A/m.

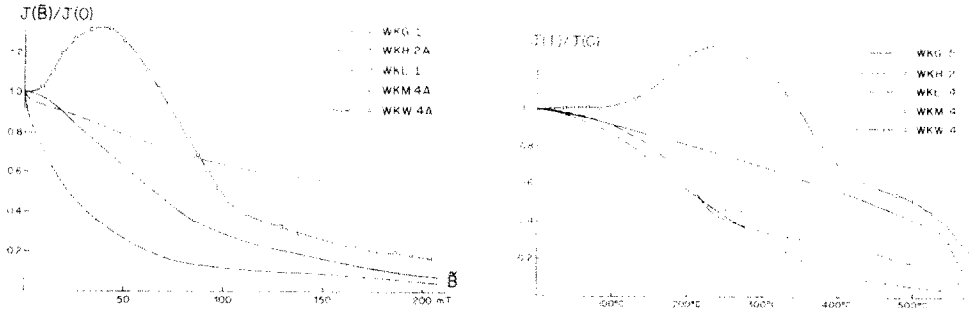


Fig. 5. Normalized decay curves of the natural remanent magnetization during progressive demagnetization with af (left) and by heating (right) of specimens from the Gypsum—Melaphyr Fm (WKW 4A and WKW 4), member 1 of the Chalus Fm (WKL 1, WKL 4, WKM 4A and WKW 4), and member 5 of the Chalus FM (WKG 1, WKG 5, WKH 2A, and WKH 2).

WKW (see Table I); the section near Lut is located in poorly exposed, steep gullies.

Our study of polished thin sections shows that the formation consists of porphyritic olivine basalts, intergranular olivine basalts, and olivine basalts rich in feldspar. The lava flows are slightly altered. Olivine occurs as phenocrysts and is found in all flows; olivine is usually altered to a mass of chlorite and serpentine, and has a rim of opaque material. In the fresh flow WKT, olivine with 85% Fo is found. Plagioclase occurs abundantly in the groundmass often as zoned and twinned laths with anorthite contents ranging between 45% and 55%. The subhedral crystals of clinopyroxene are zoned with colourless cores and lilic brown margins; they have a titano-augitic composition. The ores, chiefly titanomagnetites, are unmixed into magnetite with lamellae of ilmenite. There is no martitization; hematite occurs as rims around the crystal outlines of olivine.

MEMBER 1 OF THE CHALUS FORMATION

Pilot specimens from all ten sites — eight lava flows and two beds of tuffite — were subjected to both af demagnetization and heating. The demagnetization diagrams show that during progressive demagnetization the non-characteristic components of magnetization are removed (Fig. 6); the diagrams of the specimens of site WKO reveal that the non-characteristic components of magnetization have a direction approximately opposite to that of the characteristic remanence direction. However, site WKO contains two distinct clusters of characteristic directions with opposite polarities (Table I), the site possibly represents two separate lava flows.

The decay of remanence is illustrated in Figs. 5 and 7. The tuffites have initial remanence intensities about a factor 100 less than those of the lavas.

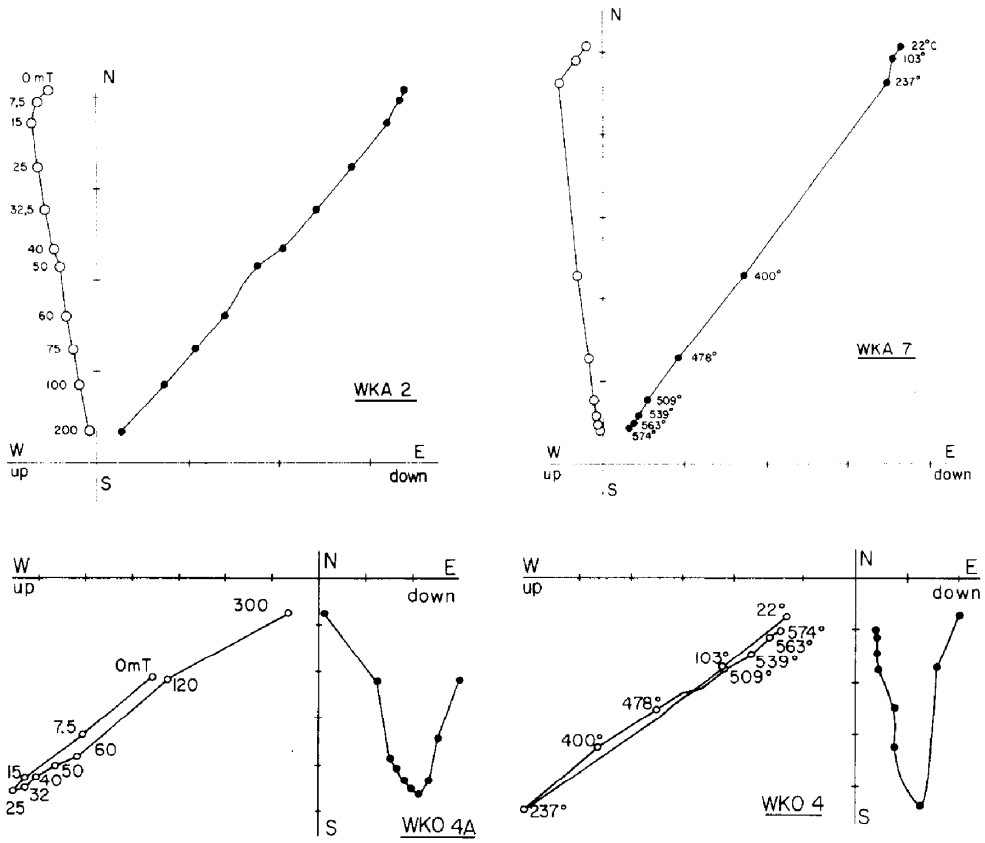


Fig. 6. Diagrams showing the progressive demagnetization of specimens from member 1 of the Chalus Formation both with af (left) and with heating (right) before tectonic correction. Each unit on both axes represents 0.5 A/m on the WKA 2 and WKA 7 diagrams and 1 A/m on the WKO 4A and WKO 4 diagrams. For further explanation see caption of Fig. 4.

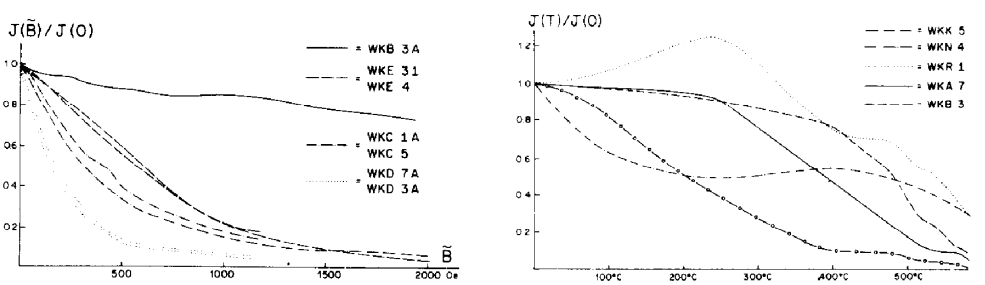


Fig. 7. Normalized alternating magnetic field (left) and thermal (right) decay curves of specimens from tuffites (WKB and WKN) and from volcanics of member 5 of the Chalus Formation.

The tuffites resist af treatment (Fig. 7: WKB 3A); after heating at 574°C no remanence is left in specimen WKB 3, but specimen WKN 4 still has 30% of the initial intensity. The deviating characteristic site mean direction of WKN is not included in the final analysis. Member 1 contains sites with normal polarity as well as sites with reversed polarity (Table I).

The lava flows of this member are mainly intergranular olivine basalts with phenocrysts of olivine in a groundmass of plagioclase, clinopyroxene, often olivine, and opaque minerals. The lavas WKL and WKK are porphyritic olivine basalts with phenocrysts of olivine, pyroxene, and opaques in a groundmass of plagioclase, clinopyroxene, rarely olivine, and opaques; the groundmass has a slightly fluidal texture.

The alteration of the lavas is moderate, but olivine is always altered to a mass of either chlorite or chrysotile; with the exception of lava WKR, the decomposed olivines are surrounded by a rim of opaques. Plagioclase occurs as small laths, usually zoned and twinned with anorthite contents ranging between 55% and 65%. Clinopyroxene occurs both as small phenocrysts and in the groundmass, and is a Ti-bearing augite. Magnetite and ilmenite are the main opaques in the intergranular rocks. The small opaque phenocrysts in the lavas WKL and WKK consists of a network of magnetite and ilmenite in which the magnetite is removed, leaving a residual network of ilmenite. Individual crystals of ilmenite are met with as well. The opaques surrounding olivine are hematite.

MEMBER 5 OF THE CHALUS FORMATION

From the six sampled lavas of member 5 the specimens of one flow — site WKF — revealed viscous magnetizations; no useful data could be obtained from this site. For the remaining sites it holds that the demagnetization results of pilot specimens from a particular flow are consistent after af treatment as well as after heating. The example of Fig. 8 shows that there are practically no secondary components of magnetization.

The decay of the remanence intensity of some specimens is illustrated in Figs. 5 and 7. The af decay curves of Fig. 7 (left diagram) show that different specimens from a particular site may have similar characteristics. After heating at 574°C the remanence intensity almost disappears (Fig. 5: WKG 1, WKH 2A).

Site WKE has two separate clusters of characteristic remanence directions, which is possibly caused by an unnoticed tilt of part of the lava flow. The lava flows of member 5 of the Chalus Formation have normal polarities.

With the exception of flow WKC, the lavas of this member are porphyritic felsic rocks with large phenocrysts of plagioclase, microphenocrysts of olivine and clinopyroxene, and rarely opaque minerals in a fine-grained groundmass of plagioclase, varying amounts of clinopyroxene, rarely olivine and opaques. Flow WKC is an intergranular olivine basalt of type as similar to that found in member 1.

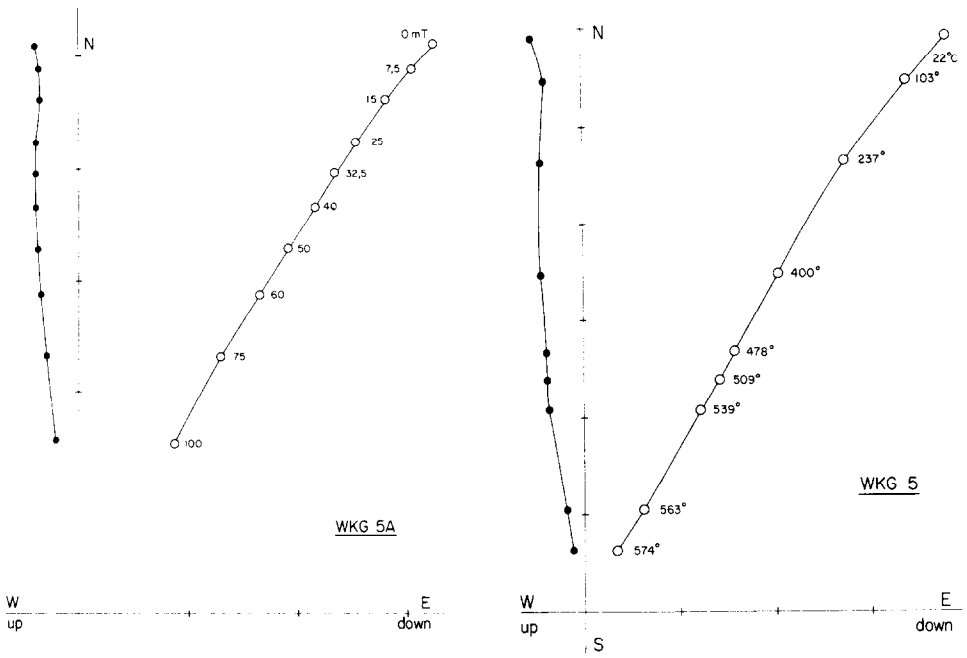


Fig. 8. Demagnetization diagrams of specimens from member 5 of the Chalus Formation with af (left) and with heating (right) before tectonic correction. Each unit on either axis represents 0.5 A/m. For further explanation, see caption of Fig. 4.

Practically all flows are slightly altered. Plagioclases occur as large aggregates and as crystals in the shape of laths. Olivine shows a strong deuteric alteration, but its crystal outlines are still observable, olivine usually has a rim of opaques. The main opaque minerals are titanomagnetites that occur both as small phenocrysts and in the groundmass. The opaque rims around olivine consists of hematite.

RESULTS AND DISCUSSION

By using progressive demagnetization procedures in our paleomagnetic investigations of the volcanic rocks of Cretaceous age from the Alborz we were able to remove the secondary components of magnetic remanence. From the decay curves (Figs. 5 and 7) one can infer that the specimens do not strongly resist af demagnetization; progressive heating shows that magnetite is probably the main carrier for the characteristic remanence, because after heating at 574°C there is little remanence left (Curie-temperature magnetite is 568°C). Inspection of the opaque minerals has confirmed that magnetite is an important constituent. It appears also that hematite, which is

TABLE II

Mean characteristic directions of magnetization of rock units of Cretaceous age from the Alborz

Locality	Rock unit	Age	E	N	D	I	k	α_{95}
Chalus Valley	Chalus formation Member 5	Coniacian Turonian	1	5	42.8	44.1	22	16.6
Chalus Valley	Chalus formation Member 1	Aptian Barremian	1	9	22.2	48.7	36	8.6
Haraz Valley	Gyps-Melaphyr Formation	pre-Aptian post-Malm	1	6	41.3	46.8	15	18.1
Central Alborz	All units	Cretaceous	—	20	33.3	47.5	22	7.2

E and N are the number of sites, excluded from and included in the ultimate analysis, respectively; D and I are the declination and inclination in degrees of the magnetization direction; k is the precision parameter; α_{95} is the semi-angle of the cone of 95% confidence, in degrees.

TABLE III

Paleomagnetic pole positions of rocks of Cretaceous age from the Alborz

Age	Position of rock unit		Pole position		σ_M	σ_P
	latitude (°N)	longitude (°E)	latitude (°N)	longitude (°E)		
Turonian—Coniacian	36.45	51.15	52.2	145	13	20.8
Barremian—Aptian	36.45	52.15	70.2	154.9	7.5	11.3
Post-Malm—pre-Aptian	36.15	52.30	54.4	143.5	15.1	23.3
Cretaceous	36.3	51.8	61.0	147.5	6.1	9.3

σ_M and σ_P are the semi-axes of the oval of 95% confidence, for the ple position.

frequently observed around the altered crystals of olivine, contributes to the remanence.

The initial NRM intensities of the lavas vary between 0.1 and 8 A/m; the tuffites have an initial NRM intensity of about 0.01 A/m. The initial Q-values -- the ratio of the intensity of remanence to that of the induced magnetization -- range from 0.3 to 34.

In Table I, the mean characteristic directions of magnetization of the individual sites are presented with their statistical data. Table II gives the mean characteristic directions of magnetization of the successive rock units as well as of the combined 20 sites included in the ultimate analysis. The paleomagnetic pole positions derived from the data of Table II are listed in Table III. Mean characteristic magnetization directions are plotted in an equal-area projection (Fig. 9).

The outflow of lavas from the Gypsum—Melaphyr Formation may have been partly or wholly synchronous with the outpourings of the lavas of member 1 of the Chalus Formation. The lavas of member 5 of the Chalus Formation are definitely 20—30 Ma younger in age.

Petrographically, the Cretaceous basaltic rocks of the Alborz belong to two distinct rock suites. The lavas of both the Gypsum—Melaphyr Fm and member 1 of the Chalus Fm are alkalic olivine basalts. The effusiva of member 5 of the Chalus Fm can be classified as "high alumina" basalts (Kuno, 1960; MacDonald and Katzura, 1964). Chemical analysis of basaltic rock samples from the Chalus area confirm the occurrence of both types of rock (Cartier, 1971).

The statistics in Table II show quite large values for the semi-angle of the cones of confidence: the characteristic magnetization directions of the sites of the respective rock units have fairly large dispersions, except in the case of member 1 of the Chalus Fm. It is possible that the number of sites included in the analyses is insufficient; it is also possible that the application of some inaccurate data for the tectonic correction is responsible for

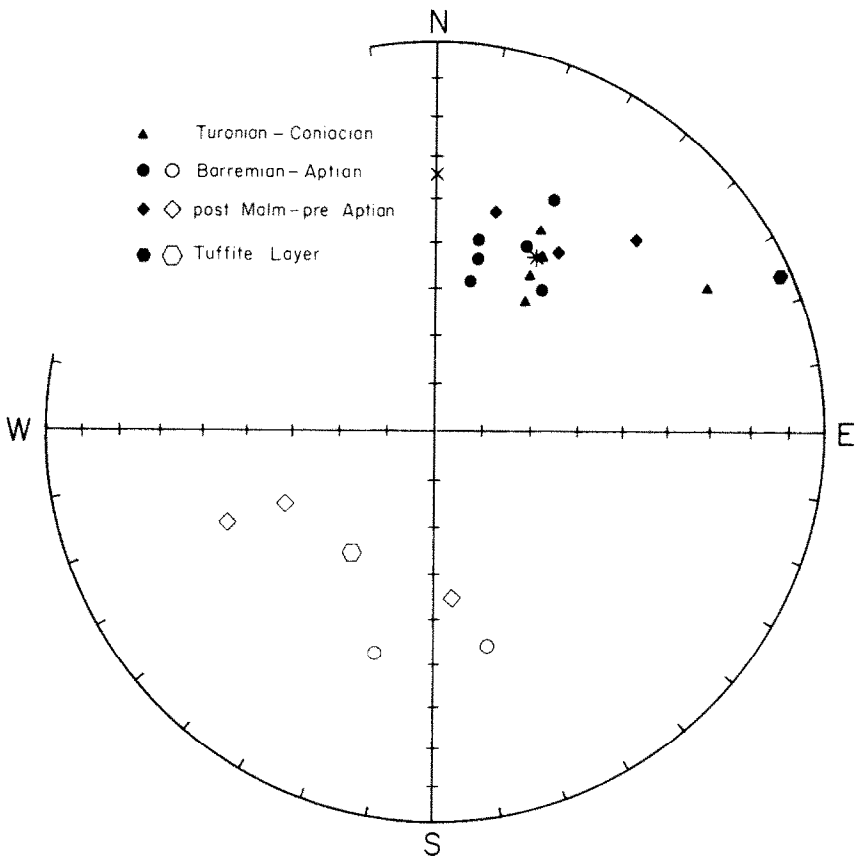


Fig. 9. Equal-area projection with the mean characteristic remanence directions of sites from the volcanics of Cretaceous age from the Alborz Mountains, Iran. Open symbols denote upward-pointing directions; closed symbols denote downward-pointing directions. The star indicates the downward-pointing site mean direction; the cross marks the downward-pointing local direction of the present axial geocentric dipole field.

the scatter in the directions of magnetization. Nevertheless, the position of the paleomagnetic pole computed from the mean of all sites fits quite well on the apparent polar wandering (APW) path of Eurasia, as designed by Irving (1977). The pole position derived from the rocks of member 1 of the Chalus Formation even coincides with the pole position 130 Ma ago on Irving's APW path. The paleomagnetic poles derived from the other rock units of Cretaceous age of the Alborz are positioned south of their respective positions of comparable age on the Eurasian APW path.

It is noticeable, however, that the paleomagnetic results obtained from rocks of Cretaceous age of the Soviet Union collected not far from the

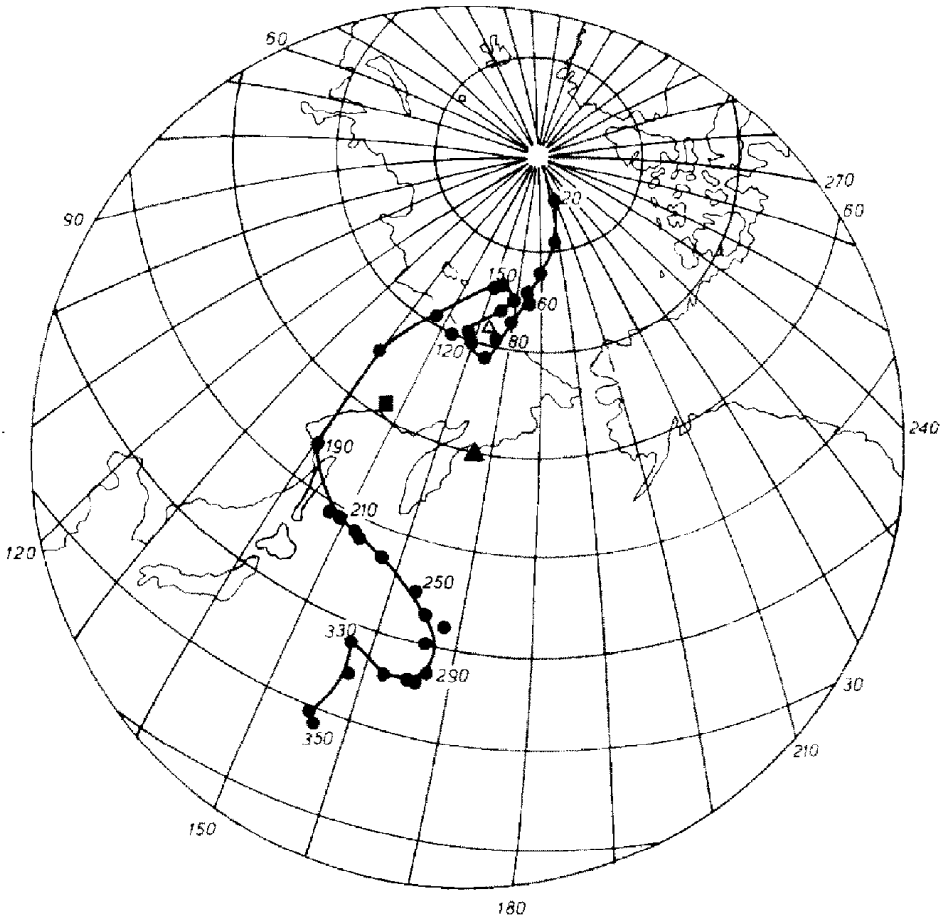


Fig. 10. Apparent polar wander path relative to northern Eurasia since the Devonian after Irving (1977) with the pole positions derived from Cretaceous rocks of the Alborz, Iran (full square), of the Hissar Range, Ouzbekistan (open triangle), and of the Bol'shoi Range, Tourkmenistan (full triangle).

Iranian border agree fairly well with our results (Khramov, 1977): rocks of Early Cretaceous age from the Hissar Range (38°N 67°E) provided a pole at 72°N 164°E; the Bol'shoi Range (39°N 55°E) showed an Early Cretaceous pole at 60°N 167°E (Fig. 10).

Our paleomagnetic data indicate that in Cretaceous times the northern part of Iran was at about its present location with respect to Eurasia. The paleomagnetic data derived from the Geirud basaltic lavas, which poured out at the Devonian—Carboniferous boundary, demonstrated that at that time northern Iran was situated at the rim of Gondwanaland (Wensink et al., 1978). Preliminary paleomagnetic results derived from volcanic rocks of

both Permian and Early Jurassic age from the Alborz Mountains indicate that Iran formed a part of Gondwanaland until post-Permian times. Iran rifted away from Gondwanaland possibly in Triassic times; subsequently, the microcontinent soon welded to Asia.

Both the Gypsum—Melaphyr Formation and the member 1 of the Chalus Fm yielded sites with normal as well as reversed magnetic polarity. During the Early Cretaceous there are epochs with normal as well as with reversed polarity. The lavas of rock unit member 5 have normal polarities only. The latter lavas poured out during the Mercanton Normal Polarity Interval or the Gubbio Long Normal Zone, which lasted from 120 Ma to 79 Ma ago. A few short-lived events with reversed polarity occur during this interval. But, from Cenomanian through Santonian times — 102–79 Ma ago - no events with reversed magnetic polarity have been observed (Van Hinte, 1976, Lowrie and Alvarez, 1977). It was during this period that the lavas of member 5 of the Chalus Fm flowed out.

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